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(54) Title: **Eco drop scheduler ai based intelligent system for predictive e waste collection schedulling and carbon-efficient resource allocation**

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(57) Abstract:

The present invention discloses an intelligent and adaptive e-waste collection scheduling and optimization system designed to improve environmental sustainability and operational efficiency through artificial intelligence-driven analytics. The system operates as a web-based platform that enables users to schedule e-waste pickups while allowing administrators to optimize logistics usmg demand prediction, hotspot clustering, route optimization, and carbon emission analysis. The invention integrates real-time geolocation processing, machine learning-based forecasting, and optimization algorithms to enhance waste collection efficiency and reduce carbon footprint. By addressing inefficiencies in manual scheduling and route planning, the system provides a scalable, secure, and environmentally responsible waste management solution suitable for municipalities, recycling organizations, and sustainability initiatives.



FORM 2
THE PATENT ACT 1970
(39 of 1970)
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The Patents Rules, 2003
~~PROVISIONAL~~/COMPLETE SPECIFICATION
(See section 10 and rule13)

1. TITLE OF THE INVENTION: Eco-Drop Scheduler:-"AI-Based Intelligent System for Predictive E-Waste Collection Scheduling and Carbon-Efficient Resource Allocation"

2. APPLICANT(S):

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3.PREAMBLE TO THE DESCRIPTION

COMPLETE

The following Specification describes the invention and the manner in which it is to be performed:

4. DESCRIPTION

The Technological Area of the Invention:

The present invention relates to the field of Environmental Technology (EnvTech) and Smart Waste Management Systems, specifically to artificial intelligence-enabled logistics optimization platforms that enhance e-waste collection scheduling, route planning, and carbon emission reduction through predictive analytics and geospatial intelligence.

Context and Thematic Exposition:

Urban waste collection systems face significant inefficiencies due to unpredictable demand patterns, scattered pickup locations, and suboptimal routing strategies. Manual scheduling often results in increased travel distance, fuel consumption, operational cost, and environmental impact.

Existing digital waste platforms primarily focus on request registration but lack intelligent demand forecasting, hotspot detection, and carbon-aware optimization. The invention addresses these limitations by introducing an AI-powered scheduling and optimization system that integrates predictive analytics, clustering, and routing intelligence into a unified platform for sustainable e-waste management.

The Invention's Integrated Ideas:

The invention introduces a data-driven and sustainability-focused architecture in which pickup operations are optimized after analyzing historical demand, geospatial distribution, and route efficiency metrics.

The system integrates user scheduling, AI-based demand prediction, hotspot clustering, route optimization, carbon tracking, and strategy recommendation modules within a single platform. Continuous optimization is achieved through daily analytics and carbon-saving evaluation, enabling adaptive operational planning and environmental impact reduction.

System Architecture:

The Eco Drop Scheduler system architecture comprises a multi-layered intelligent e-waste collection platform integrating a user interface layer, an AI-enabled backend processing layer, centralized data storage, administrative analytics interfaces, and external intelligent services. As illustrated in the architecture diagram, the system enables bidirectional data flow between user requests, predictive analytics, optimization engines, and administrative decision-support modules to achieve efficient and sustainable e-waste collection operations.

FastAPI-Based AI Backend Processing Layer:

The core intelligence of the invention resides in a FastAPI-based backend that orchestrates predictive analytics, clustering, routing optimization, and strategy generation. This layer processes incoming requests from the user module, interacts with the centralized database, and delivers optimized operational insights to the administrative interface. The backend

integrates multiple AI engines including demand prediction, hotspot clustering, route optimization, carbon analysis, and large-language-model-based strategy recommendation, forming the central decision-making unit of the system.

Centralized Data Storage (Supabase Database):

All operational and analytical data are stored in a centralized cloud database comprising structured tables including e-waste request records, carbon optimization history, user profiles, reward data, and role management. The database supports real-time retrieval and update operations required for predictive modeling, clustering analysis, and route optimization. Persistent storage of historical pickup and emission data enables continuous learning, performance tracking, and long-term sustainability analytics across collection cycles.

AI Demand Prediction Engine

The demand prediction module applies machine learning techniques to historical pickup request data stored in the database to forecast temporal and spatial demand patterns. By estimating future request density across regions and time windows, the module enables proactive vehicle allocation, scheduling preparation, and resource planning. Predicted demand information is transmitted to the administrative dashboard to support strategic operational decisions.

Hotspot Clustering Engine:

The hotspot detection component employs K-Means clustering to group geographically proximate pickup requests into high-density zones. This spatial aggregation reduces redundant travel paths and supports region-wise collection planning. Identified clusters are visualized within the administrative interface and utilized by the routing engine to construct optimized collection routes covering multiple nearby requests efficiently.

Route Optimization and Carbon Analysis Engine:

The route optimization engine computes efficient pickup paths using nearest-neighbor and distance-minimization algorithms based on clustered locations and geocoded coordinates obtained via external mapping services. The engine generates optimized routes that reduce total travel distance, fuel consumption, and operational time. A carbon optimization module estimates emissions for both naive and optimized routes and calculates carbon savings

metrics, which are stored in the carbon history database and presented in the administrative analytics dashboard.

AI Strategy Recommendation Module (LLM-Based):

The system integrates a large language model-based strategy engine that analyzes outputs from prediction, clustering, and carbon optimization modules to generate high-level operational recommendations. These include vehicle deployment strategies, emission reduction measures, and efficiency improvements. The module interacts with external AI services to provide adaptive decision support to administrators for sustainable logistics planning.

Administrative Analytics and Control Interface (Admin End):

The administrative layer provides a dashboard for monitoring demand forecasts, hotspot clusters, optimized routes, carbon metrics, and user requests. Administrators can approve or reject pickup requests, review operational analytics, and evaluate sustainability performance indicators. This interface transforms AI outputs into actionable operational controls, enabling data-driven management of e-waste collection systems.

External Service Integration Layer:

The system integrates external services including geocoding APIs for converting addresses into geographic coordinates and large language model services for strategic analytics generation. These integrations enhance spatial accuracy and intelligent decision support capabilities without increasing system complexity.

Newness and Inventive Step:

The invention presents a novel integration of demand prediction, geospatial clustering, route optimization, and carbon emission analytics within a unified e-waste scheduling platform. Unlike conventional waste management systems that rely on static scheduling or manual planning, the present invention introduces adaptive AI-driven optimization and sustainability metrics, constituting a significant inventive step in intelligent environmental logistics systems

Use in Industry:

The invention is applicable across municipalities, recycling agencies, smart city initiatives, environmental organizations, and corporate sustainability programs. It supports scalable e-waste collection operations, improves logistics efficiency, and enables measurable carbon reduction in waste management workflows.

Advantages:

- ✓ Enables intelligent e-waste pickup scheduling and optimization
- ✓ Reduces travel distance and operational cost
- ✓ Minimizes carbon emissions through optimized routing
- ✓ Identifies high-demand collection zones using clustering
- ✓ Supports data-driven sustainability planning
- ✓ Provides scalable web-based deployment

Eco Drop Scheduler

AI-Powered E-Waste Collection System

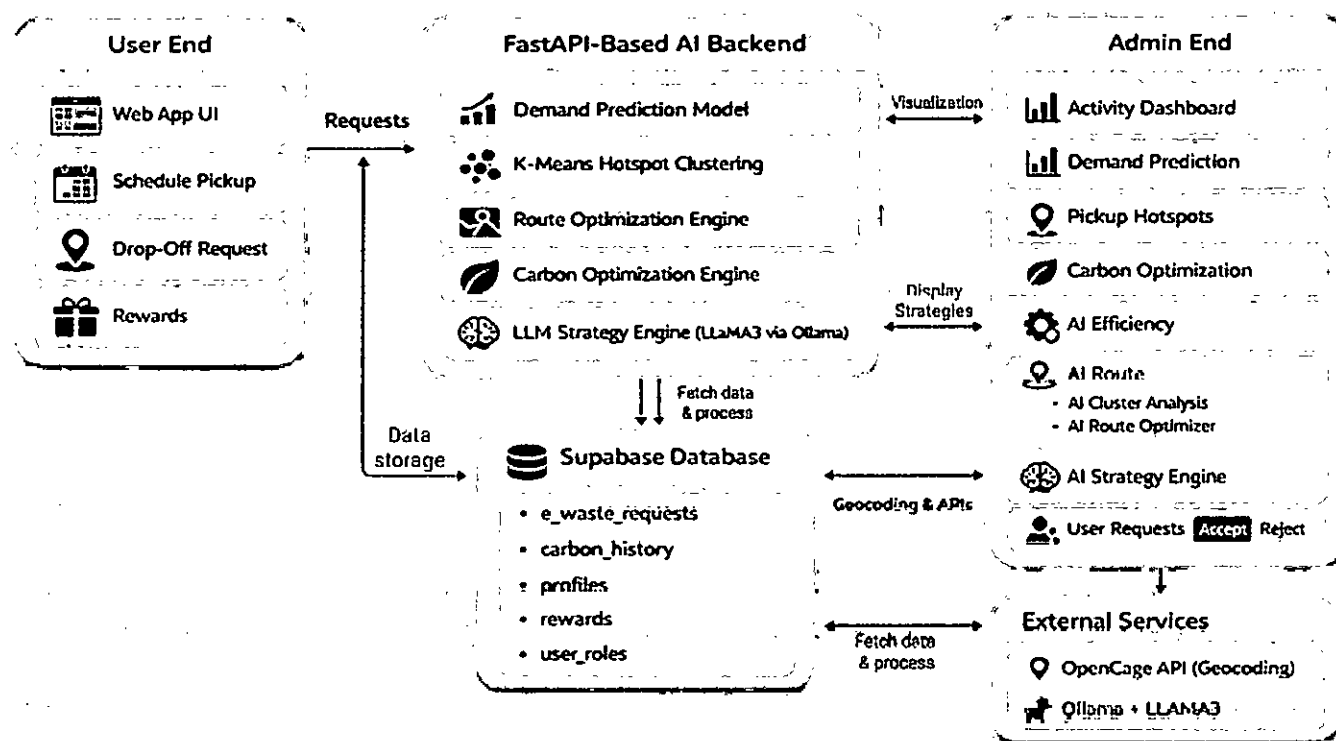
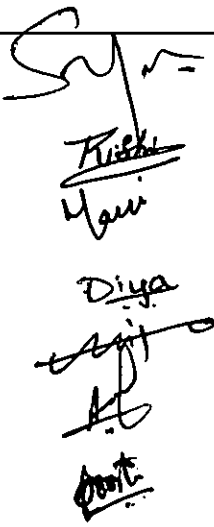


Figure 1: Architectural Diagram

5. CLAIMS

1. We claim, the system operates as an intelligent e-waste collection scheduling platform comprising a user request module, an AI-driven demand prediction engine, hotspot clustering, and route optimization framework, producing optimized pickup schedules and reduced-emission collection routes through a unified web-based application environment.
2. We claim, the intelligent waste management system includes a centralized database storing pickup requests, geolocation coordinates, scheduling data, and carbon emission metrics for efficient analytics and operational optimization.
3. We claim, the optimization system integrates real-time geocoding, clustering analysis, carbon emission estimation, and adaptive strategy generation to improve environmental sustainability and operational efficiency without manual route planning.

6. DATE AND SIGNATURE

Jain (Deemed-To-Be University), Rishin Aiyappa A G, Dr. Manivasagam G, Diya A Bhandari, Mit Desai, Ahmed Ali Ahmed, Spoorti M Mattur	
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7. ABSTRACT OF THE INVENTION

The present invention discloses an intelligent and adaptive e-waste collection scheduling and optimization system designed to improve environmental sustainability and operational efficiency through artificial intelligence-driven analytics. The system operates as a web-based platform that enables users to schedule e-waste pickups while allowing administrators to optimize logistics using demand prediction, hotspot clustering, route optimization, and carbon emission analysis. The invention integrates real-time geolocation processing, machine learning-based forecasting, and optimization algorithms to enhance waste collection efficiency and reduce carbon footprint. By addressing inefficiencies in manual scheduling and route planning, the system provides a scalable, secure, and environmentally responsible waste management solution suitable for municipalities, recycling organizations, and sustainability initiatives.